CENTER FOR HISTORY OF CHEMISTRY

VLADIMIR PRELOG

Transcript of an Interview
Conducted by

Tonja Koeppel

in

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on

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Vladimir Prelog JH 3/15/96

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INTERVIEW: Vladimir Prelog

INTERVIEWED BY: Tonja Koeppel

PLACE: Swiss Federal Institute of Technology
Zürich, Switzerland

DATE: 17 January 1984

KOEPPEL: Professor Prelog, you were born in Sarajevo on July, 23, 1906. To what country did Sarajevo belong at that time?

PRELOG: It was part of the Austro-Hungarian monarchy. My life has been affected in many ways because Sarajevo has such a bad reputation. As you may know, Archduke Franz Ferdinand and his wife were assassinated there in 1914 and that action is considered to be the proximate cause of the first World War.

KOEPPEL: Yes.

PRELOG: I realized how Sarajevo's reputation affected me when I had to acquire a sailing permit in 1950 in the United States. While doing so, an agent of the Internal Revenue Service interviewed me. He asked me, "Professor, where are you from?"

I said, "Zürich."

He responded, "Zürich, Sweden?"

I said, "No, Zürich, Switzerland."

He was a little irritated and said to me, "You were not born, of course, in Zürich."

I answered, "No."

He asked, "Where?"

I answered, "Sarajevo."

He then said, "Oh, that's the place where all this mess started."

I was very proud that while he didn't know that Zürich is in Switzerland, he did know that Sarajevo exists.

KOEPPEL: He didn't hold it against you, did he?

PRELOG: No.

KOEPPEL: That was in 1950?

PRELOG: In 1950. Sarajevo is now famous because the Winter Olympics will be held there this year.

KOEPPEL: Yes. Well, you must have been a student in Sarajevo when the assassination occurred. Were you eight years old?

PRELOG: I was a student in elementary school at the time and

I was standing very near to the place of assassination. As a pupil I was assigned to throw flowers at the Archduke's procession. I was in the first row.

KOEPPEL: Oh, you were there!

PRELOG: The assassination occurred about two hundred meters from the place where I was standing. I didn't see it, however, because it happened around the corner.

KOEPPEL: It must have been terrible.

PRELOG: Yes, it was very bad.

KOEPPEL: There was much confusion, I guess.

PRELOG: I was, as in many other cases, a spectator of very important events.

KOEPPEL: Oh, yes, you were indeed! Two world wars. We'll get to that later. So, you went to grade school in Sarajevo.

PRELOG: Yes. We then moved to another place, Zagreb. During most of my stay in Yugoslavia I resided in the city of Zagreb, the capital of Croatia, the northern part of Yugoslavia.

KOEPPEL: What year did you move to Zagreb?

PRELOG: In 1915.

KOEPPEL: So you were in about the third grade. You entered a different school and then...

PRELOG: I then attended Realgymnasium, a kind of high school, something between elementary school and university.

KOEPPEL: Oh, I see.

PRELOG: The system is practically the same in Switzerland.

KOEPPEL: Does it take you to the university level?

PRELOG: You attend Realgymnasium for eight years and then take what we call the maturity examination. After passing this examination you may be accepted by a university. This is more or less the middle European system.

KOEPPEL: Yes. You must have studied chemistry there because I noticed that you published a paper while still in this school.

PRELOG: For two years I attended school in a smaller town, Osijek, because my father was appointed director of a high school there. I had a very, very good teacher. I helped him

to build a small chemistry laboratory. As I have mentioned on several occasions, I proved that the level of chemical literature was not very high because my paper was accepted! It was published when I was about fifteen years old.

KOEPPEL: That's wonderful! What was the paper about?

PRELOG: It was about a titration apparatus.

KOEPPEL: You actually wrote it in German!

PRELOG: I wrote it in German and it was corrected by my teacher. It was accepted by the Chemiker Zeitung.*

KOEPPEL: You were bilingual?

PRELOG: No, I could only write German, not speak it. I had studied German at school. My mother tongue was Croatian and we didn't speak German at home.

KOEPPEL: I see.

PRELOG: We read German books at home. There were many German books at home, but we didn't speak German.

KOEPPEL: Yes, well, what was your father's name?

PRELOG: My father's name was Milan. It is a Yugoslav name. He was a professor at the Gymnasium. There were different branches of Gymnasia. One emphasized Greek and Latin for people who wanted to become lawyers or even study medicine. The other branch offered mathematics, natural sciences, and modern languages. It was called Realgymnasium.

KOEPPEL: I see.

PRELOG: I had taken the scientific education, not the humanistic one.

KOEPPEL: Your father was not a scientist?

PRELOG: My father was a historian at the Gymnasium. Very soon, however, he became professor of history at the University of Zagreb. We moved back therefore to Zagreb. The last years of my schooling at this level were in Zagreb.

KOEPPEL: Did your father's being an historian influence you? You seem to have an historical bent, at least in your latest work.

PRELOG: Yes. I come from a family of historians--even my

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*Vladimir Prelog, "Eine Titriervorrichtung," Chemiker Zeitung, 45 (1921): 736.

grandfather was a historian. I have a brother who is an historian, and my son is an historian. My son is now at the University of Marburg.

KOEPPEL: How did you become a chemist? Was your mother scientific in any way? Did she have a science education?

PRELOG: No, but I was always very much interested in science and I think perhaps I became a chemist as a kind of protest against my father.

KOEPPEL: Yes, that could very well be; it happens very often. So, you then moved to Prague to study chemical engineering.

PRELOG: Yes. At that time, you know, chemical engineering was very different from what it is understood to be today. It was really chemical technology. There was an institute of chemical technology in Prague. It was called a technical high school (technische Hochschule) as were many similar schools in Germany, but it had university status. It was not a high school in the American sense.

KOEPPEL: No.

PRELOG: Technische Hochschule is translated as institute of technology. For example, the Eidgenössische Technische Hochschule is the Swiss Federal Institute of Technology. It's somewhat like MIT. The school that I attended was called the School of Chemical Engineering. It was not engineering as it is taught today, which places very strong emphasis on apparatus and technical processes. It was partly pure science and partly descriptive technology. It was for a long time the same here in Zürich. It was not chemical engineering.

KOEPPEL: Yes, I was a student at the ETH and I remember that. So, you started to study chemistry in Prague. You described this in one of your recent publications--or actually I think it was a lecture, "Thoughts After 118 Semesters of Chemistry."*

PRELOG: Yes. I don't think, however, that American schools use the semester.

KOEPPEL: Yes, they do. In a way, though, you're right; many universities go on trimesters. The conventional way, however, is still the semester.

PRELOG: It is fifty-nine years now since I started at the Institute of Technology in Prague.

KOEPPEL: What you want to say is that you were always a

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*Vladimir Prelog, "Gedanken nach 118 Semestern Chemiestudium," Chemie und Gesellschaft, G. Boche ed., (Stuttgart: Wissenschaftliche Verlagsgesellschaft, 1984).

KOEPPEL: What was wrong with his formula?

PRELOG: Oh, the size of the rings was not so easily determined. He put a five-membered ring in place of a six-membered ring. Our first discovery was that this ring can't be a five-membered ring.

KOEPPEL: Well, you already had a lot of experience with polycyclic compounds. Did you synthesize adamantane?

PRELOG: Yes. That was more or less a secondary research activity, but we had in fact been the first to synthesize adamantane. By doing so we also proved its structure. The structure of adamantane had been known, but it had been more or less a matter of intuition, supported by a Debye-Scherrer X-ray diagram that didn't really prove the structure. It showed only that the proposed formula was compatible with the X-ray diagram.

KOEPPEL: Was your formulation two-dimensional, or did you propose a three-dimensional structure?

PRELOG: Not I, but Lukes was really the one who first proposed the right formula, more or less intuitively, because the crystals of adamantane are tetrahedral. He proposed it because a hydrocarbon, $C_{10}H_{16}$, with a melting point of about 280°C, was quite unusual. This is possible only because the molecule is almost perfectly spherical, and therefore, fits the lattice of the crystal very well. The other point was the tetrahedral crystal structure. From those two bits of knowledge and by using intuition, he proposed the structure. The powder diagrams didn't specify the position of the individual atoms in this case. Therefore, our synthesis represented a synthetic structure proof. We wrote only two papers about adamantane, but they made quite an impression.*

KOEPPEL: You did that before you came to Zürich.

PRELOG: Yes, that happened in Zagreb.

KOEPPEL: Could you tell me a little bit about Woodward. You published several papers with him.**

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*Vladimir Prelog and Rativoj Seiwerth, "Über die Synthese des Adamantans," Berichte der Deutschen Chemischen Gesellschaft, 74B (1941): 1644-48; and Prelog and Seiwerth, "Über eine neue, ergiebigere Darstellung des Adamantans," ibid., 74B (1941): 1769-72.

**D.H.R. Barton, O. Jeger, V. Prelog, and R.B. Woodward, "The Constitution of Cevine and Some Related Alkaloids," Experientia 10 (1954): 81-90; F. Gautschi, Jeger, Prelog, and Woodward, "Über die Konstitution der Decevinsaure," Helvetica Chimica Acta, 37 (1954): 2280-2294; Jeger, Mirza, Prelog,

KOEPPEL: You may be a little bit in limbo, because you say that you forgot your mother language and then maybe you didn't get quite as deeply into your next languages.

PRELOG: Yes, you know, either you have to learn a language very young, or you have to be very talented.

KOEPPEL: I know you don't have a crystal ball, but what is your idea of the future of chemistry or of chemical education?

PRELOG: Chemistry is a very big field and it's so important because everything that happens in the world is related to it. It is so important to know in order to understand what's going on in the world. So, we have really just started and I think...

KOEPPEL: It will survive?

PRELOG: It will never stop. Consider, for example, such a complicated process like memory. Evidently, memory is something chemical. Yet we are far from understanding it. We have to do a lot of fundamental work to understand it. We don't have the necessary tools yet, but they do improve every day. Of course, Einstein said, "We live in a time of more and more perfect methods and more and more confused aims." We always have to think of what we really want to do because the possibilities now are really tremendous.

KOEPPEL: Yes, they are frightening.

PRELOG: If you take a chain of one hundred amino acids in a protein, and twenty different amino acids, you can make 10^{100} possible combinations. There are, as far as we know, 10^{86} atoms in the world and many, many proteins have hundreds or thousands of amino acids. Now, we know that we will never be able to make all of these possible isomeric proteins even though we could synthesize many of them with existing methods. If you say I want to combine a hundred amino acids in a certain sequence to a protein, you can do it. It will take years of effort, however, and require several very able coworkers. Yet, if you fail to put only a very few amino acids into the right sequence, you will not have the expected activity. Which one of these syntheses should be done? Should we spend a million dollars to prepare one of these isomers which might be of no value in the end? This is our problem. We have to be clear about the aim, about what we really want to do because we can now do much more than we were able to do before. We improve our methods every day. They become easier, faster, and more specific.

KOEPPEL: Professor Prelog, before we conclude the interview, I would like to ask whether you have any letters, papers, or

manuscripts on file. Where will your papers be stored? Are you collecting them?

PRELOG: Unfortunately, I have too many of them. It would be very difficult to sort out what is important and what is not important. I think that I shall keep the correspondence with Cahn, Ingold, and other people who contributed to the CIP system.

KOEPPEL: Where are these letters located? Are they kept right at the Institute?

PRELOG: I have them here now, but I shall perhaps deposit them into the archive of the library of the Swiss Federal Institute of Technology. The library does quite a good job collecting the correspondence and other papers of some of the members of the faculty. The library, for example, will keep all of the important correspondence that Ruzicka left. It has a special place where it keeps the correspondence and manuscripts of Thomas Mann, who was a kind of professor here during his later years. It also has the correspondence of Carl Gustav Jung and some other important members of the ETH. I think the library's staff does it professionally and very well. I don't think that I have many letters worth preserving. Some of them, of course, are more important than others, but I think that it would be a formidable task to select them and to store them. I think that I will not leave too much of the correspondence to the archives of the Swiss Federal Institute.

KOEPPEL: You will donate only what is relevant to your scientific work.

PRELOG: Only very relevant. Otherwise, very soon even the Federal Institute of Technology wouldn't have enough space.

KOEPPEL: Professor Prelog, I thank you for the interview you gave me. You have given me so much interesting information about your life, about your work and your relationship with scientists, including those in America. It has been a great experience. Thank you very much.

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